Machine Learning based Software Trigger for Downstream Tracks

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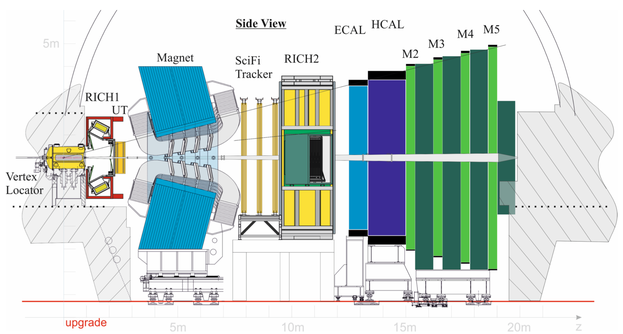
Abstract

Long-lived Particles(LLPs) are produced within the detector raising the challenges to identify and track the particles. Downstream Tracks, the tracks that are generated from the decay of LLPs shows signal profile outside Vertex Locator(Velo) leaving signals in Upstream Tracker and SciFi Tracker. The primary objective of the research is to build a new sophisticated software trigger system for the detection of these Downstream Tracks. The research has two major milestones. The first one is to design and deploy a machine learning-based trigger in finding the Seed Tracks from the Ghost Tracks and the second bit is to design a model to cherry-pick Downstream Tracks from the set of Tracks after the initial process of Track Reconstruction. Advanced Neural Network Architecture is designed to replace TMVA based Multi-Layer Perceptron Model. This fine-tuned model will get deployed later on in Track Reconstruction Sequence after multiple iterations of testing.

LHCb Experiment



Seed-Ghost Classifier

In context of LHCb experiment, Downstream Tracks are produced from the decay of Long-lived Particles(LLPs). LLPs decay outside VeLo detector showing signal profiles only in UT and SciFi Trackers. For the reconstruction and tracking designated track reconstruction algorithms are designed, it starts with taking track segments from SciFi Trackers called Seed-Tracks and extrapolate them to respective UT signals.

In the first phase of research, the primary goal is to identify and tag the real seed-tracks and enhance the seed-track quality for subsequent track reconstruction. The experimental data shows non-true tracks, produced from combinatorials, called Ghost Tracks.

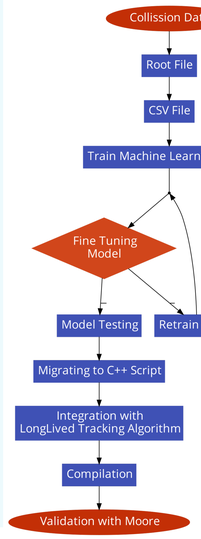
To address this issue, A boosting based ML algorithm has developed that is trained on Run3 simulation data to specifically to eliminate and reduce the Ghost Tracks. Using CatBoost, a novel Machine Learning frame work has demonstrated significant overall improvements in the data quality by keeping the efficiency almost unchanged. The development of the model from data to integration has met various challenges including the complexity of combinatorial tracks, optimisation of threshold parameters, minimise efficiency drop, etc.

Effective deployment of this ML-based robust software trigger shows significant improvement in data quality. By increasing the data quality of seed tracks, will lead to improved track reconstruction by folds.

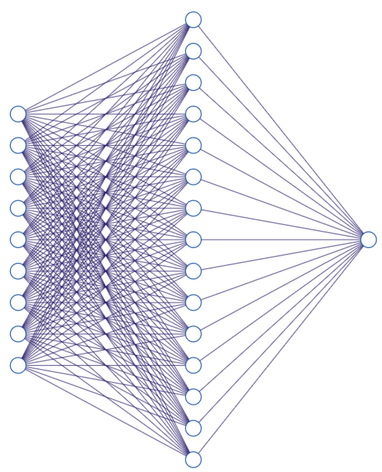
Integration and Test Pipeline

Long-lived Particle tracking is done using the designated Track Reconstruction Algorithms(TRAs). Moore, a LHCb Application designed for the Trigger Analysis helps with the analysis and developments of algorithms.

Recent versions of Moore is designed for Run3 analysis. The developments starts with finding the right simulation data for the analysis and use Monitoring Algorithms to dump the tracks to ROOT files for the flexibility and convenience. The ROOT files are then transformed to Parquet Files, a modified version of comma separated files for crunching the data volume. After picking the right kinematical variables, the model was trained and picked the optimal model configuration from the post-training analysis and multiple iterative tests. These models are then translated to C++ Scripts for the easy integration with the Downstream Tracking Algorithm. Threshold, the parameter decides the probability to be True-Ghost Tracks are then decided with the cost-analysis with efficiency and purity of the TRAs.



Track Classifier - Shallow to Deep Learning



Next phase of research involves the development and to design an algorithm to cherry-pick the Downstream Tracks. To accomplish this, we currently have a shallow TMVA-based 9x15x1 fully-connected network integrated into the reconstruction process.

This phase is characterized by two objectives,The first objective is to migrate existing TMVA-based network to a PyTorch based network, with a subsequent evaluation of its performance. The aim is to leverage the capabilities and flexibilities of PyTorch Framework to potentially enhance and for the comparitive studies.

The Second bit is to develop a more sophisticated Deep Learning Neural Network that can replace and and outperform the existing model. The design of the novel-architecture is in it's initial phase, showing promising results in it's testing and primary analysis.

Results

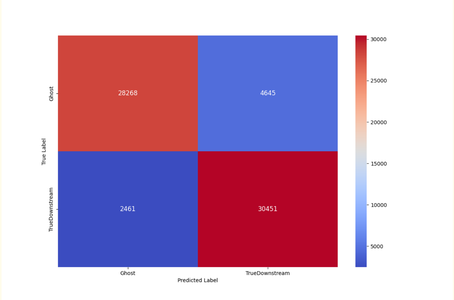


Figure 1. Confusion Matrics from Seed-Ghost Classifier

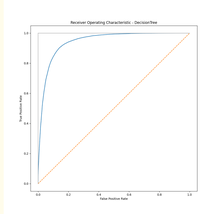


Figure 2. ROC-Curve for Seed-Ghost Classifier

References

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Frontiers in Big Data, 2022.

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